

Optimisation of Rheological Properties of Water-Based Mud (WBM) with Natural Additives by using Response Surface Methodology (RSM)

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Abstract

The rheological properties of drilling muds are critical for achieving optimal performances during drilling operations. In this study, bentonite, tannin, and xanthan gum were utilised as water-based drilling mud additives to enhance the rheological properties. The Response Surface Methodology (RSM) with Box-Behnken Design (BBD) was used to investigate the additive's effect on the rheological properties of the drilling muds. The concentration of bentonite, tannin and xanthan gum were considered as the independent variables, while plastic viscosity (PV), apparent viscosity (AV), and yield point (YP) as the responses in the design of experiment (DOE). The YP, AV and PV were determined using 9.0 ppg of drilling mud according to the API standard procedures. Response surface plots (3D) were used to analyse the effect of the independent factors on the rheological properties and resulting in R^2 values of 0.9753 for PV, 0.9582 for AV, and 0.9513 for YP, which indicates that the interactions between elements in the system were statistically significant as these R^2 values were close to 1.0. Bentonite significantly increased the PV, AV, and YP, whereas it decreased as tannin concentration increased. The optimal rheological properties required for low PV and AV with a high YP could be achieved using WBM formulation of bentonite at 4.02 g, 7.29 g of tannin, and 0.53 g of xanthan gum. Meanwhile, xanthan gum had an insignificant effect on the PV, AV, and YP. This finding demonstrates that the RSM model is an accurate and relevant tool; hence it may be utilised to optimise the experimental conditions of mud formulation and accurately predict the rheology parameters of drilling muds.

Article Info

Article history:

Received date: 01 August 2021

Accepted date: 13 September 2021

Keywords:

Water-based mud
Natural additives
Biopolymer
Rheological properties
Response surface
methodology (RSM)
Box-Behnken design
(BBD)

1.0 Introduction

The increasing number of drilling activities throughout the year has increased the demand for drilling muds. In Malaysia, hundreds of exploration wells have been drilled, and large amounts of seismic data have been acquired to understand more about the country's basins. A total of 163 oil fields and 216 gas fields were discovered in this region throughout the exploration drilling by the Production Sharing Contracts (PSC) [1]. Proper selection of drilling mud is a critical aspect in drilling phase realisation. Additionally, it is included in designing drilling mud to meet the requirement of the well. Various drilling mud functions can only be achieved when mud properties, including pH, density, rheology, and filtration, are maintained and optimised. By optimising these

properties, any difficult drilling conditions could be avoided [2].

Drilling muds were used and expected to improve the productivity of the drilling activities. Drilling muds are generally classified into three major groups; water-based mud (WBM), oil-based mud (OBM) and synthetic-based mud (SBM) [3]. WBM is a complex mixture containing water as a continuous phase with other additives to achieve specific physical properties. Meanwhile, OBM and SBM exhibited high-performance fluids in thermal stability, high ROP, and lubricity [4], [5]. However, OBM and SBM usage in the industries has been limited due to environmental concerns during post-drilling activities [6], [7]. Thus, WBM is preferable as drilling muds in many drilling activities over OBM and SBM [8].

WBM refers to a drilling mud that is made of water,

solids, and heavy minerals. Solids containing clays and organic colloids were mixed into the mud design to attain the optimum rheological properties. Bentonite is one of the clay types that is frequently used in the formulation of drilling muds due to its desirable properties of rheological, plastic viscosity, gel strength, surface area, and cation exchange capacity (CEC) [9]–[11]. In addition, bentonite may improve the rheological properties of WBM due to its ability to absorb the water and swell properly in it [12].

The rheological properties of muds can also be modified by adding other additives into the continuous phase. Various additives, including viscosifier, weighting agent, deflocculant, fluid loss and pH control, are primarily used in oil and gas drilling [13]–[15]. Over the past few years, environmentally friendly and cost-effective additives have gained attention of researchers and industry practitioners [16], [17].

Natural additives such as biopolymers and starches improve mud rheology and minimise the filtrate volume to prevent formation damage [18]. For viscosity enhancement, a biopolymeric additive called xanthan gum (XG) was added to WBM [19]. The addition of XG also resulted in fluid loss declination due to its shear-thinning and thixotropic properties in the mud [20]. A study has been conducted by R. Da Luz et al. [21] to evaluate the impact of xanthan gum (XG) on the rheological properties and filtration control of WBM. It was found that the addition of XG acted as a better viscosifier and displayed a non-Newtonian and pseudo-plastic behaviour. Furthermore, the pseudo-plasticity of the solution in this study improves the mud's ability to hold the cuttings in the wellbore during the drilling operations, explaining the xanthan gum's role as a natural viscosifier in WBM's system [22].

In addition, another natural additive called tannin was used as a deflocculant in the WBM formulation. Deflocculant, also called thinner, was mixed into the drilling mud to avoid the mud from flocculating together [23]. Adding the thinner can reduce the mud's probability of swell, hydrate, and disperse when exposed to harsh conditions and unconventional reservoirs [12]. A study by A.H.M. Muhayyidin [24] revealed that a tannin derived from *Rhizophora spp.* was assessed as a deflocculant in various WBM blends. The result portrayed that the optimal amount of tannin (6g) could minimise the filter cake thickness and fluid loss. Therefore, the presence of tannin was perceived to serve as deflocculant in the WBM system.

The rheological properties of the WBMs were

usually tested by using Fann-35A viscometer following API standards. However, laboratory measurement is seen as a time-consuming and costly process, and it is sometimes unavailable. Hence, by accurately predicting the outcomes using an ideal method such as response surface methodology (RSM), it may effectively be assessing and optimising the rheological properties required to carry out the drilling operation [25], [26].

Menezes et al. [2] used RSM to study the influence of CMC on the rheological properties of bentonite dispersions for water-based mud. Alhajabdalla et al. [27] applied RSM and Box-Behnken Design (BBD) to optimise the stability of fibrous dispersion used in drilling and completion operations. Finally, Imtiaz et al. [28] investigated and optimised the rheological and filtration behaviour of water-based mud using RSM.

Thus, this study is aimed to determine the effect of various bentonite, tannin, and xanthan gum concentrations on the rheological properties of WBM. Response surface methodology (RSM) was used to optimise mud formulation for maximising the yield point while minimising the viscosity. The rheological properties were evaluated and analysed using the standard temperature and pressure conditions according to API standards.

2.0 Methodology

2.1 Materials

Soda ash (sodium carbonate, Na_2CO_3) and caustic soda (sodium hydroxide, NaOH) were supplied by Classic Chemicals Sdn Bhd, while xanthan gum was procured from Chemiz (M) Sdn Bhd. Bentonite and barite were both supplied by Baker Hughes Services Sdn. Bhd. Tannin used in this study was extracted from the bark of *Rhizophora spp.*, which was collected from Kuala Sepetang, Perak [29]. All the chemicals and additives used in this study were used without any further purification or processing.

2.2 Design of Experiments (DOE) and Response Surface Methodology (RSM.)

Design of experiments (DOE) is a statistical tool that allows for the control of the experiment's parameters, while response surface methodology (RSM) is a combination of mathematical and statistical techniques used to optimise the existing procedures [30]. The flowchart depicted in Figure 1 summarises the overall structure of this study. The composition of each WBM sample was initially suggested by the RSM

Table 1: Data from WBM rheological measurements

Run	Bentonite	Tannin	Xanthan Gum	PV	AV	YP
	g	g	g	cP	cP	lb/100ft ²
1	2	6	2	8	16	16
2	20	2	1	20	50	60
3	11	2	2	18	48	60
4	11	2	0	18	50	64
5	11	10	2	14	31	34
6	2	2	1	12	23	22
7	11	6	1	16	45	58
8	20	6	2	20	42	64
9	11	6	1	16	46	60
10	2	10	1	2	13	22
11	2	6	0	6	15	18
12	20	10	1	15	46	62
13	20	6	0	8	16	16
14	11	6	1	16	44	56
15	11	10	0	14	30	32

features provided by Design-Expert 13 (Stat-Ease Inc., Minneapolis, MN, USA). As can be seen in Table 1, a total of 15 experiments were composed of 16 cubes of factorial points, 4 centre points, and 8 axial points with an alpha of 2 using Box-Behnken Design (BBD). The primary advantage of the RSM-BBD methodology over other approaches is the capability to analyse several parameters with fewer experimental trials [27], [31]. Variance analysis (ANOVA) evaluated three parameters to determine the significant interaction effects on the response variables. The accuracy of the fitted polynomial model was shown by the values of R^2 and adjusted R^2 coefficient. The significant model terms were evaluated by the probability value (p-value) at a 95% confidence interval. Three-dimensional surface plots and contour plots were simulated to determine the effects of the independent variables on the response.

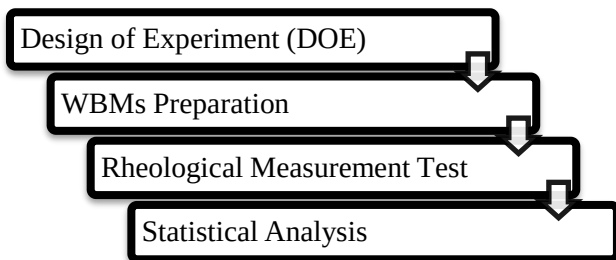


Figure 1: Methodology Flowchart

2.3 Water-Based Mud (WBM) Preparation

WBMs were prepared using the following ingredient: water as the based fluid, bentonite, caustic soda, soda ash, barite, xanthan gum, and tannin as deflocculant. All components were mixed in accordance with API standards [32]. Initially, 350 mL

of distilled water was added into the mixer cup, followed by 0.25g of soda ash and 0.25g of caustic soda, and homogeneously mixed for 4 minutes using a Hamilton Beach mixer. Soda ash was used to maintain the pH of WBM, whereas caustic soda was primarily added to reduce water hardness [33], [34]. The pH of the water was maintained at 10. Following that, different ranges of concentrations of bentonite (2-20g) and xanthan gum (0-2g) were added to the mixture according to the DOE's recommended concentration. Barite of 25g was then added and mixed for 5 minutes. To ensure complete hydration, the mud was left at room temperature to prehydrate for 24 hours. Once completed, tannin between 2g to 10g was mixed with the mixture for 5 minutes before analysing the rheological properties. All the experiments were conducted at a constant room temperature of 25°C. The composition of WBM and their mixing order are listed in Table 2.

Table 2: WBM Composition

Mix order	Additives	Mass (g)	Time (min)
1	Water	350	
2	Soda Ash	0.25	2
3	Caustic Soda	0.25	2
4	Bentonite	2-20	5
6	Xanthan Gum	0-2	15
5	Barite	25	5
7	Tannin	2-10	10

2.4 Rheological Properties Measurements

All muds were stirred for ten minutes to ensure consistent conditions before the rheological analysis. The rheological properties of each WBM sample were

Table 3: Summarised statistical analysis for responses model

Source	Sum of Squares	df	Mean Square	F-value	p-value (<0.05)	R ²	Adj. R ²	Pred. R ²
Plastic Viscosity (PV)								
Model	2088.6	1	2088.6					
Linear	18.75	3	6.25	0.2137	0.8848	0.0551	-0.2026	-0.8084
2FI	1.25	3	0.4167	0.0104	0.9984	0.0588	-0.6472	-3.1078
Quadratic	311.98	3	103.99	61.78	0.0002	0.9753	0.9308	0.7121
Cubic	5.75	3	1.92	1.44	0.4353	0.9922	0.9452	
Residual	2.67	2	1.33					
Total	2429.00	15	161.93					
Apparent Viscosity (AV)								
Model	11760.00	1	11760.00					
Linear	144.25	3	48.08	0.2320	0.8722	0.0595	-0.1970	-0.5333
2FI	37.25	3	12.42	0.0443	0.9867	0.0749	-0.6190	-1.9052
Quadratic	2206.58	3	735.53	102.39	< 0.0001	0.9852	0.9585	0.8348
Cubic	23.25	3	7.75	1.22	0.4792	0.9948	0.9634	
Residual	12.67	2	6.33					
Total	14184.00	15	945.60					
Yield Point (YP)								
Model	15746.40	1	15746.40					
Linear	297.00	3	99.00	0.2448	0.8633	0.0626	-0.1931	-0.4522
2FI	198.00	3	66.00	0.1242	0.9432	0.1043	-0.5675	-1.4350
Quadratic	4019.60	3	1339.87	29.00	0.0014	0.9513	0.8637	0.2907
Cubic	207.00	3	69.00	5.75	0.1517	0.9949	0.9646	
Residual	24.00	2	12.00					
Total	20492.00	15	1366.13					

determined using a Fann-35A viscometer, and the dial readings were recorded at various speeds (3, 6, 100, 200, 300 and 600 rpm) [32]. The plastic viscosity (PV), apparent viscosity (AV), and yield point (YP) of the WBM were determined by using equations (1), (2), and (3).

$$PV = \theta_{600} - \theta_{300} \quad (1)$$

$$AV = \frac{\theta_{600}}{2} \quad (2)$$

$$YP = \theta_{300} - PV \quad (3)$$

The value of θ_{600} and θ_{300} denote dial readings of viscometer at 600 and 300 rpm, respectively. PV and AV values were expressed in (cP), while YP values were expressed in (lb/100ft²).

3.0 Results and discussion

3.1 Mathematical Model Regression

In this study, RSM software generates regression model equations that correlate the relationship between the rheological properties of the WBMs in response to the concentration of the additives. The regression model equations are presented in equations (4), (5), and (6).

$$PV = 16.00 + 3.12A - 2.88B + 0.5C + 1.25AB + 0.0AC + 0.0BC - 5.87A^2 + 2.13B^2 - 2.13C^2 \quad (4)$$

$$AV = 45.00 + 10.87A - 6.38B + 3.25C + 1.5AB + 6.25AC + 0.75BC - 14.75A^2 + 2.75B^2 - 8.0C^2 \quad (5)$$

$$YP = 58.00 + 15.50A - 7.0B + 5.50C + 0.5AB + 12.5AC + 1.5BC - 17.75A^2 + 1.25B^2 - 11.75C^2 \quad (6)$$

Where:

PV : Plastic Viscosity A : Bentonite (g)

AV : Apparent Viscosity B : Tannin (g)

YP : Yield Point C : Xanthan Gum (g)

By using the above equations, the predicted values of the responses can be obtained. These equations have the same positive sign of coefficients, which indicates a positive effect, whereas if the coefficients show a negative sign, it means a negative impact [35][36]. In simple words, coefficients with positive signs had positive impacts, while those with negative signs had

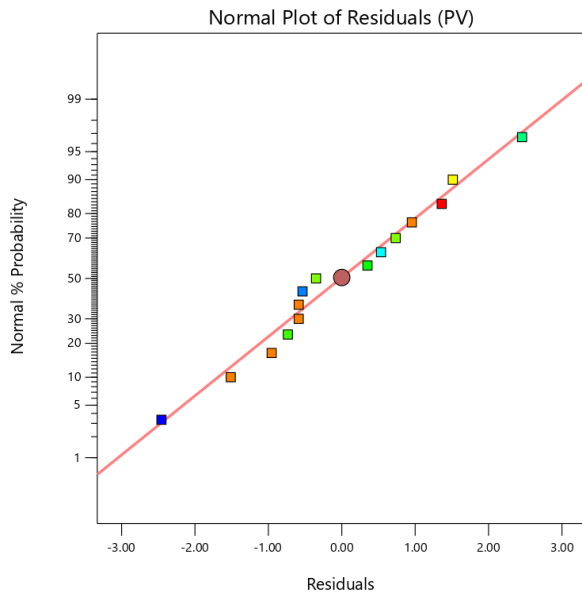


Figure 2 (a): Normal Plot of Residuals (PV)

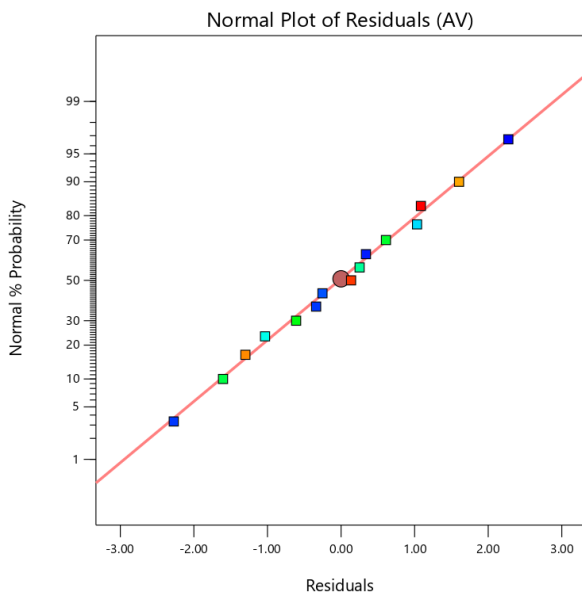


Figure 2 (b): Normal Plot of Residuals (AV)

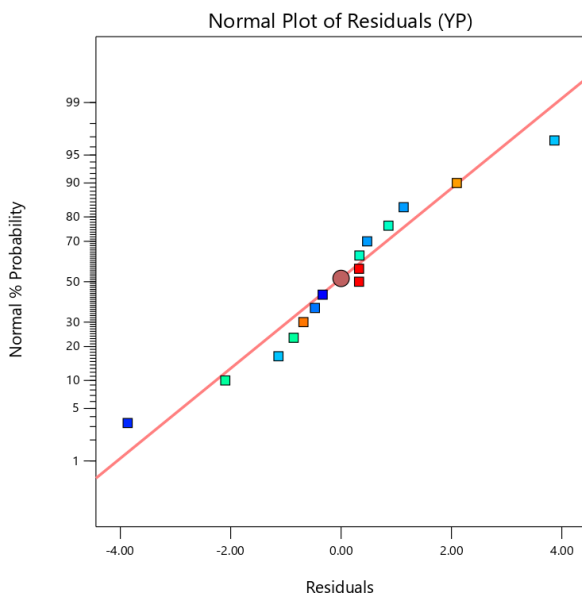


Figure 2 (c): Normal Plot of Residuals (YP)

negative effects. For example, in Eq. (4), the coefficient of 16.00, an independent value from any factor, shows a positive sign; the same goes for the linear terms of A, C, interaction terms of AB, and second-order terms B. In contrast, the other terms in the equation show negative signs. Hence, it can be summarised that PV was increased as these positive terms increased while decreased if the negative terms increased.

3.2 ANOVA Analysis of Rheological Properties

In this study, analysis of variance (ANOVA) based on BBD was conducted using Design-Expert Software Version 13.0 to assess the model's significance and fitness. In addition, ANOVA was used to statistically correlate the relationship between the individual input factors (bentonite concentration, tannin concentration, xanthan gum concentration) with the output responses (PV, AV, and YP).

To determine the effectiveness of the regression model towards the responses, the probability P-value model must be less than 0.05 to be considered significant, as shown in Table 3. A study conducted by Mishra et al. has shown experimentally that the model to optimise the preparation of activated carbon from *Limonia acidissima* shell was found statistically significant as the P-values were less than 0.05 [31]. Besides, the value of R^2 also need to be closed to unity (1.0) so that the model can be considered reliable, and the lack of fit must be non-significant as the model needs to fit [28]. AV was observed to have the highest value of R^2 (0.9852), followed by PV (0.9753) and YP (0.9513). Referring to the criteria mentioned earlier, each quadratic model of the responses was confirmed as the perfect regression model for this study. All of these models have shown significant R^2 value and insignificant P-value of lack of fit.

As can be seen, the errors on the residuals' normal probability plot in Figure 2 (a, b, and c) are typically distributed along a straight line and were insignificant. Besides, as all of the data points in each normal plot lie on or near the straight lines, it may be assumed that the data was distributed regularly in all of the responses. The fitted line in this plot validated the significance of the p-value mentioned in the ANOVA analysis tables, thus confirmed the model's adequacy.

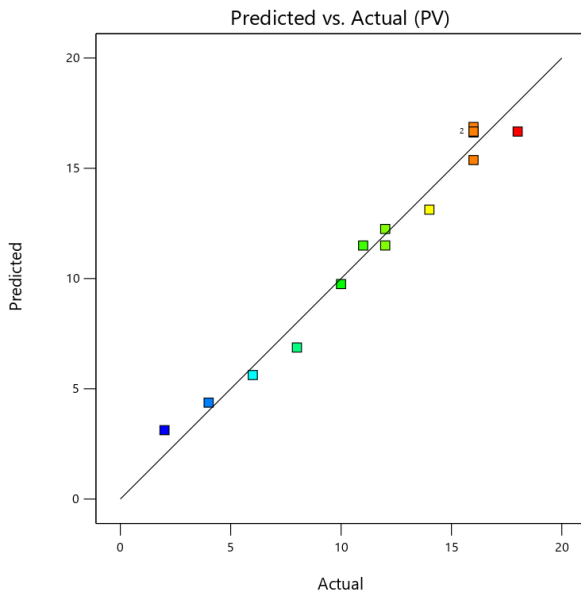


Figure 3 (a): Predicted vs Actual for PV (cP)

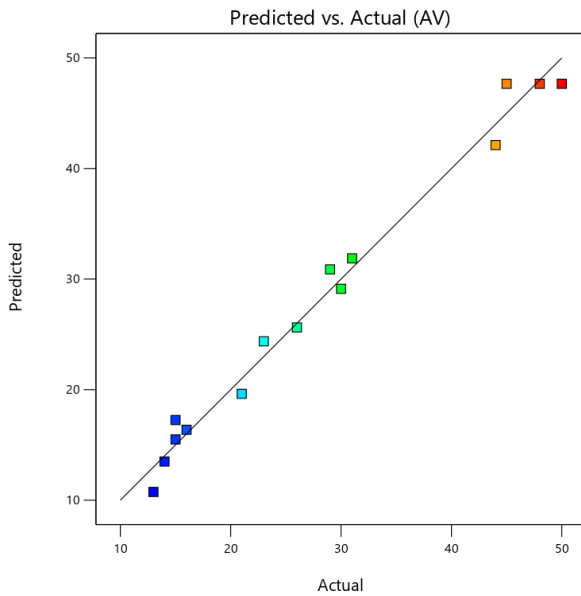


Figure 3 (b): Predicted vs Actual for AV (cP)

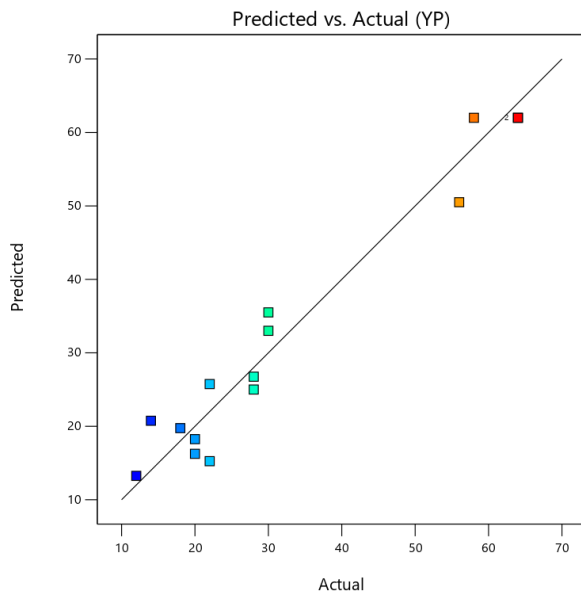


Figure 3 (c): Predicted vs Actual for YP (lb/100ft²)

Figures 3 (a, b, and c) provide the predicted vs actual plots for PV, AV, and YP responses. All the data points showed similar good distribution along the straight line, implying an excellent agreement between the predicted and experimental response values. Thus, all of these statistical tests demonstrated that the quadratic models provided by the software were effective in correlating the relationship between process factors and responses. Furthermore, the fitted data points along the straight line in these three plots synchronised with the high value of adjusted R^2 for all three responses implied a high degree of correlation between the actual and predicted values [31].

3.3 Interaction Effect of Process Variables on Responses

The 3D plots described by the regression model were constructed to interpret the response surfaces between the factors (concentrations of bentonite, tannin, and xanthan gum) with the responses (PV, AV and YP) of the water-based mud.

3.3.1 Influence of bentonite and tannin concentrations (g)

Figure 4 (a and b) show the relationship between bentonite concentration and tannin concentration towards the responses of PV and YP of the WBMs. As shown in both figures, bentonite increases the PV and YP. This observation is confirmed by a study from V. Matoh [37], where they employed that the rheological behaviour of water-based mud was increased with increasing bentonite concentrations. Meanwhile, it was found that the increment in concentration of tannin decreased the PV and YP. A study by K. S. Hafshejani et al. [23] revealed that the rheological parameters of bentonite-based mud reduce drastically after being mixed with tannin, which acts as an excellent deflocculant. Deflocculant reduces the PV and YP of the mud by providing an alkaline environment that neutralises the positive edge of clay layers and results in the clay particles repelling one another, thus avoiding them from flocculating together [23].

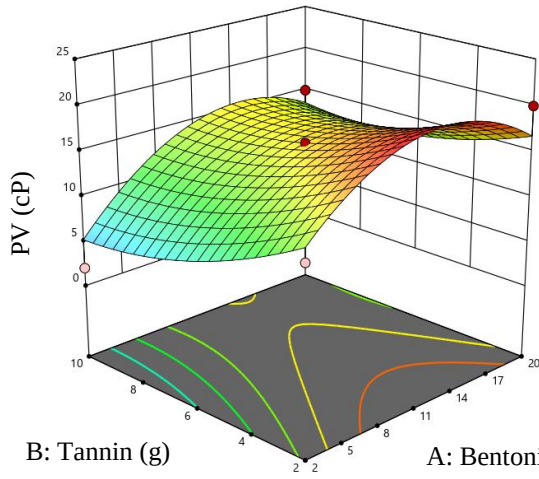


Figure 4 (a): PV response for tannin concentration versus bentonite concentration

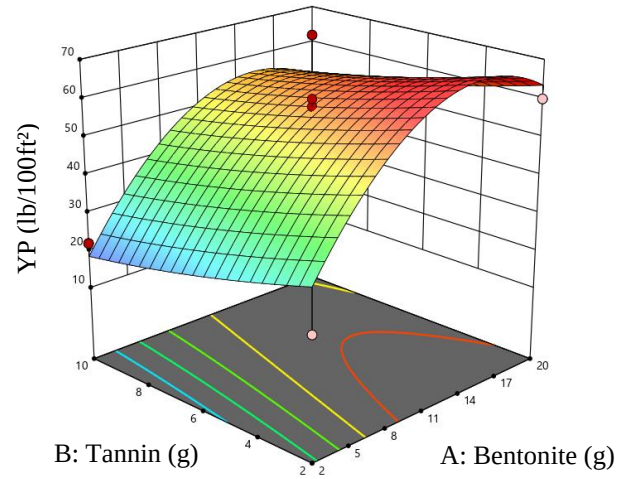


Figure 4 (b): YP response for tannin concentration versus bentonite concentration

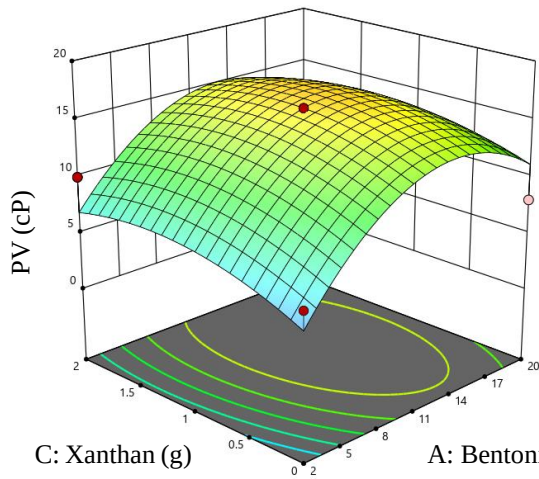


Figure 5 (a): PV response for xanthan gum concentration versus bentonite concentration

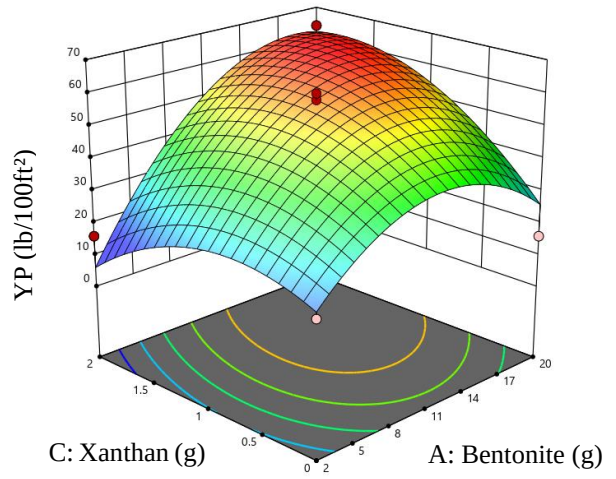


Figure 5 (b): YP response for xanthan gum concentration versus bentonite concentration

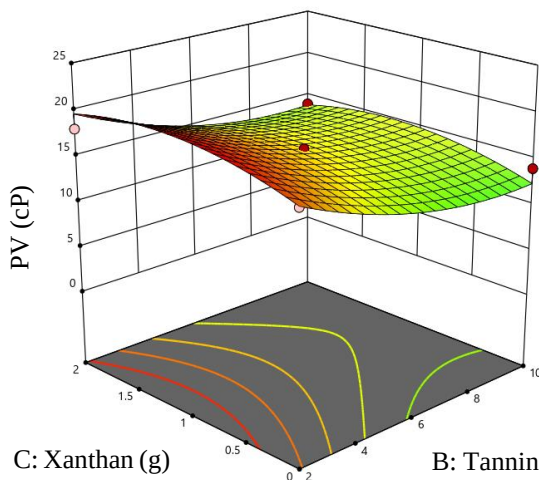


Figure 6 (a): PV response for xanthan gum concentration versus tannin concentration

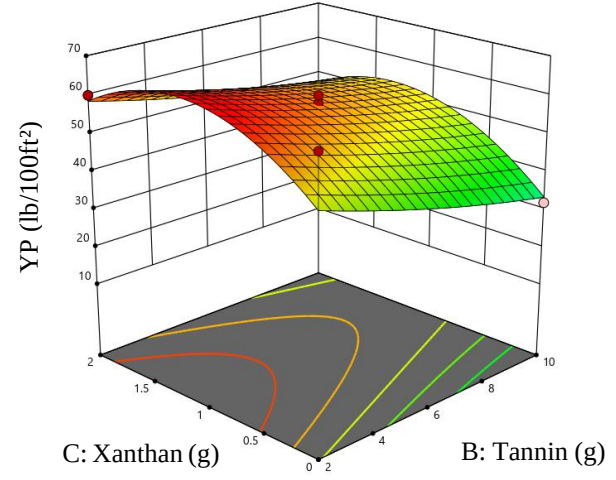


Figure 6 (b): YP response for xanthan gum concentration versus tannin concentration

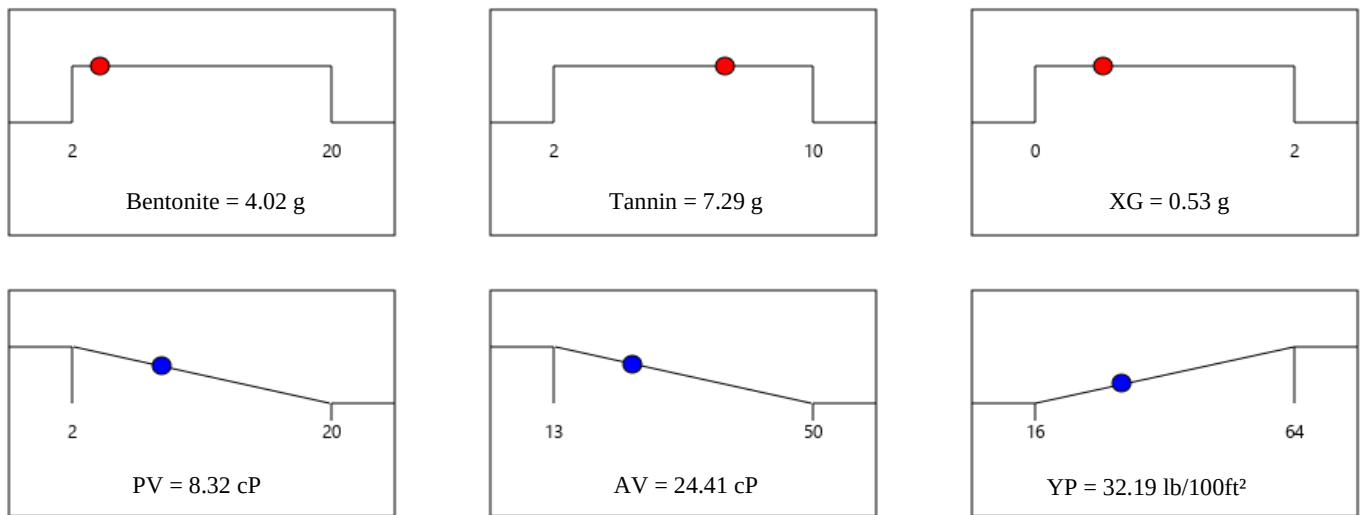


Figure 7: Ramp blocks representing optimised process condition

3.3.2 Influence of bentonite and xanthan gum concentrations (g)

Figure 5 (a) observed that both bentonite and xanthan gum increased the plastic viscosity (PV). Both additives acted as a viscosifier in the drilling mud systems, thus explaining the significance of PV improvement in this 3D model. Similar trends have been observed in Figure 5 (b) for bentonite and xanthan gum concentrations against yield point (YP). The mixing of bentonite and xanthan gum shows more significant improvement in this YP model compared to the previous PV model (Figure 5(a)) due to the behaviour of the xanthan gum, which improves the YP while gradually increase the viscosity [38]. A high YP/PV ratio is crucial because it gives a flatter velocity profile, which is desirable for mud to efficiently transport the cuttings [14]. Additionally, the addition of xanthan gum in the formulation also significantly affected the rheological properties due to its shear-thinning and thixotropic properties in the mud [39]. A similar observation has been reported by J.W. Powell et al. [40], where the usage of xanthan gum drastically improved the rheology and fluid performances. The addition of both bentonite and xanthan gum is statistically significant, thus the response surfaces of the PV and YP for given tannin values resulted in being curved, which enhanced the PV and YP of the mud.

3.3.3 Influence of tannin and xanthan gum concentrations (g)

Figure 6 (a) found that the concentration of tannin (2-10g) decreased the PV value gradually, while vice versa for xanthan gum concentration. The combination of both natural additives does not strongly affect the 3D

plot as the combination of tannin and xanthan gum did not enhance the PV property. Figure 6 (b) suggests that by considering the tannin concentration at a constant value of 2g, the YP increases with increasing xanthan gum concentration until it reaches a maximum value at 64 cP and starting to decrease once it reaches the maximum point. This observation was also confirmed by Akpan et al. [38], who studied the xanthan gum's influence on the rheological properties of drilling muds. This lower concentration of xanthan gum could enhance the YP of the drilling mud. When mixed with a small amount of xanthan gum, the clay particles can improve the mud suspension. However, at higher xanthan gum concentrations, both xanthan gum and clay particles agglomerated together and lost their capacity to improve the mud suspension. Altogether, the YP of WBM was lowered gradually in the presence of medium to high concentrations of xanthan gum.

3.4 Process Optimisation

In preparation of water-based mud (WBM), relatively low viscosities are desired since it helps to improve the rate of penetration (ROP). At the same time, it needs to be high enough for the YP to lift the cuttings out of the hole, thus reducing the probability of having associated drilling problems during drilling operations. In certain conditions, such as during drilling the HPHT wellbore, deflocculants are needed to prevent the WBM from dehydrated due to thermal degradation. Therefore, deflocculant is used to reduce the PV, AV and YP of the WBM whenever appropriate and required, especially when drilling HPHT wellbore. However, optimising the responses under the same conditions proved challenging due to their difference in behaviour towards the rheological properties. For example, bentonite and xanthan gum improved the

rheological properties, whereas tannin reduced it. Hence, it is decided that the best possible goal is to get the optimum point for each factor by minimising PV and AV response while maximising the YP.

The parameters of the additives were adjusted to be in the range the same as previous experiments. As a result, the optimised values for the factors were as follows: bentonite = 4.02 g, tannin = 7.29 g, and xanthan gum = 0.53 g to achieve optimum PV of 8.32 cP, AV of 24.41 cP and YP of 32.19 lb/100ft² as shown in Figure 7.

Table 4: Comparison between predicted vs experimental value of rheological properties at optimum conditions

Factors	Predicted Value	Experimental Value
PV (cP)	8.32	8.67 ± 0.58
AV (cP)	24.41	24.33 ± 0.58
YP (lb/100ft ²)	32.19	31.33 ± 1.15

Validation experiments were carried out in triplicate using the optimal conditions generated from the software. Table 4 shows the predicted and experimental results of the response at optimum conditions. In total, the p-value of the validation run is below 0.05, which employed the experimental value found to agree well with the model-derived values. As a result, RSM demonstrated its efficacy in optimising experimental conditions.

4.0 Conclusions

In this study, the optimisation of rheological properties of the water-based drilling mud was conducted based on RSM using the Box-Behnken Design (BBD) method. The purpose was to enhance the WBM performances in drilling operations. The effect of bentonite, tannin, and xanthan gum concentrations on the rheological properties was investigated using Fann-35A viscometer, and the data generated were then analysed using ANOVA and response surface plots (3D). From the observation, bentonite and xanthan gum increased the rheological properties, while tannin decreased it. In addition, it was found that the interaction between xanthan gum and bentonite towards the rheological properties of WBM is the most significant. The ability of the clay structure to associate together to improve the rheological and fluid performance is the essential factor in interactions between the variables. All responses show that the results obtained fitted the quadratic models, indicating that the software effectively correlated the relationship between process factors and responses. The optimum

conditions of WBM formulation were obtained at 4.02 g of bentonite, 7.29 g of tannin, and 0.53 g of xanthan gum. The optimised rheological properties have a plastic viscosity about 8.32 (cP), apparent viscosity about 24.41 (cP), and yield point about 32.19 (lb/100ft²). Validation experiments were done using the optimised WBM formulation and the results obtained show good agreement with the predicted values. Considering the significance of the statistical results in this study, it can be concluded that the RSM models are adequate in predicting and optimising the rheological parameters of WBM with high accuracy and reliability.

Acknowledgement

The authors extend our sincerest gratitude to Universiti Teknologi MARA, Shah Alam, for providing us with the infrastructure and instrument facility to carry out the research work.

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